# LECTURE 03 - TOPOLOGICAL PROPERTIES OF LIE GROUPS, HAAR MEASURE

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In this lecture (and most likely the next), we will give a crashed summary on some foundational results on Lie groups and their Lie algebras. Lots of theorems are based on differential geometry (real or complex), hence will be take for granted. As we will completely ignore these proofs, we will not restrict ourselves to rigorous logic, meaning that we summarize results based on contents, not on a logical order of proofs. Students are encouraged to take the notes as a *dictionary*. When you accumulate enough experience, you are encouraged to study the proofs (even after this course).

## 1. Preliminaries on Topological Groups

**Definition 1.1.** *A* ***topological group*** *$G$ is a topological space as well as a group such that the map*

$$G \times G \to G, \qquad (x, y) \mapsto xy^{-1}$$

*is continuous. Here as usual, $G \times G$ is equipped with the product topology.*

**Example 1.2.** *Equipped with the Euclidean topology, $\mathbb{R}^n$, $\mathbb{C}^n$ are topological groups under usual addition.*

**Example 1.3.** *Many function spaces, such as $L^2(\mathbb{R}^n)$, are topological vector spaces. Thus, they are topological groups under usual addition (in those function spaces). Hmm....., of course, one may ask, for instance, a good definition of $L^2(\mathbb{R}^n)$. We will discuss this after introducing Haar measure.*

**Example 1.4.** *Equip $\mathrm{Mat}_n(\mathbb{R})$ and $\mathrm{Mat}_n(\mathbb{C})$ with the usual Euclidean topology. Then equip the group $\mathrm{GL}_n(\mathbb{R})$ (resp. $\mathrm{GL}_n(\mathbb{C})$) with the subspace topology so that it becomes an open subset in $\mathrm{Mat}_n(\mathbb{R})$ (resp. $\mathrm{Mat}_n(\mathbb{C})$). As multiplication and inverse are polynomials of matrix entries, they are continuous. In this way, $\mathrm{GL}_n(\mathbb{R})$ and $\mathrm{GL}_n(\mathbb{C})$ are topological groups.*

The following are two remarks concerning the Hausdorfness in the definition of topological groups.

**Remark 1.5.** *Hausdorfness are usually taken by many but not all mathematicians in the definition of topological groups (and we will do so after Theorem 1.16). This is because (at least) occasionally people may encounter non-Hausdorff topological vector spaces in research.*

**Remark 1.6.** *Some students may be familiar with algebraic groups. One may think that non-Hausdorfness is allowed in the definition of topological group because one may wish to include algebraic groups, such as* $\mathrm{GL}_n(\mathbb{R})$ *(under Zariski topology), into the category of topological groups. Yet this is not the case. In fact, algebraic groups are not topological groups. The definition of algebraic group* $G$ *requires* $G \times G \to G, (x, y) \mapsto xy^{-1}$ *to be a morphism of varieties, hence the topology on* $G \times G$ *is not the product topology.*

**Definition 1.7.** *Let* $G, H$ *be topological groups. We say* $f : G \to H$ *is a* ***homomorphism of topological groups*** *if it is continuous and is a homomorphism of abstract groups. If* $f$ *is bijective and has a continuous inverse, then we say it is an* ***isomorphism****.*

The following example shows that a bijective homomorphism of topological groups may not be an isomorphism.

**Example 1.8.** *When the group* $\mathbb{R}$ *is equipped with the discrete topology, the so-obtained topological group is denoted by* $\mathbb{R}_{dis}$*. Then the identity map* $\mathbb{R}_{dis} \to \mathbb{R}$ *is a bijective homomorphism, but not an isomorphism.*

With some mild additional assumptions on topological groups, we can claim that a bijective homomorphism is an isomorphism, as the following theorem claims.

**Theorem 1.9.** *Let* $G$ *and* $H$ *be separable locally compact topological groups, and* $f : G \to H$ *be a bijective homomorphism. Then* $f$ *is an isomorphism.*

*Proof.* A proof is included in the Appendix. □

**Definition 1.10.** *Let* $G$ *be a topological group. By a* ***topological subgroup****, we mean a pair* $(\iota, H)$*, where* $H$ *is a topological group and* $\iota : H \to G$ *is a continuous injection.*

**Remark 1.11.** *The above definition emphasizes possible different embeddings of small groups into big groups. For instance, it is ambiguous if we say* $\mathrm{SL}_2(\mathbb{R})$ *is a subgroup of* $\mathrm{Sp}_4(\mathbb{R})$*, see Example 3.11 for details. Yet if there is no ambiguity (for instance* $H$ *is already a subset of* $G$*), then we just say* $H$ *is a subgroup of* $G$ *for simplicity.*

Now we aim to explain why Hausdorfness assumption is very mild in the definition of topological groups. We need the following two useful lemmas. As their proofs are quite standard (and short) when dealing with topological groups, we also include them.

**Lemma 1.12.** *Let* $G$ *be a topological group and* $H$ *be a subgroup. Then* $\overline{H}$ *is a closed subgroup.*

*Proof.* Take $x, y \in \overline{H}$. We need to check that $xy^{-1} \in \overline{H}$. In fact, the continuity of the map $(x, y) \mapsto xy^{-1}$ implies that for every neighbourhood $U$ of $xy^{-1}$, we can find an open neighbourhood $V_x$ of $x$ and an open neighbourhood $V_y$ of $y$ such that $V_x \cdot V_y^{-1} \subseteq U$. Since $V_x \cap H$ and $V_y \cap H$ are nonempty, $U \cap H$ is nonempty. Thus $xy^{-1} \in \overline{H}$. □

**Lemma 1.13.** *For every open neighbourhood $U$ of $e$, there exists an open neighbourhood $W$ of $e$ such that $W \cdot W^{-1} \subseteq U$. This implies that there always exists a closed neighbourhood $K$ of $e$ contained in $U$ that is symmetric, i.e. $K = K^{-1}$.*

*Proof.* The existence of $W$ follows directly from the continuity of the map $(x, y) \mapsto xy^{-1}$. We claim that $\overline{W}$ is a closed neighbourhood of $e$ contained in $U$. For every $x \in \overline{W}$, $xW \cap W$ is nonempty. Hence $x \in W \cdot W^{-1} \subseteq U$. Now take $K := \overline{W} \cap \overline{W}^{-1}$. Then $K$ is a closed and symmetric neighbourhood (since it contains the open neighbourhood $W \cap W^{-1}$), and $K \subseteq \overline{W} \subseteq U$. □

Based on the above two lemmas, we have the following corollary.

**Corollary 1.14.** *In a topological group $G$, the closure of $\{e\}$*

(1) *is the intersection of all neighbourhoods of $e$ in $G$;*
(2) *is the smallest closed subgroup of $G$ containing $e$;*
(3) *is normal.*

*Proof.* See the Appendix. □

**Remark 1.15.** *If a topological group $G$ satisfies the $T_0$ axiom, then one can prove that it is a $T_1$-space by symmetry. In fact a topological group $G$ satisfying $T_0$- (hence $T_1$-) axiom is regular (hence Hausdorff, see the next theorem). This is why in most scenarios, people assume a topological group to be Hausdorff.*

**Theorem 1.16** ([Mi-To91, Chapter 1 Theorem 1.6])**.** *Let $G$ be a topological group (not necessarily Hausdorff) and $H$ be a subgroup.*

(1) *Then the canonical projection map $\pi : G \to G/H$ is continuous and open.*
(2) *If $H$ is further assumed to be closed, the $G/H$ is regular hence Hausdorff. In particular, when $G$ is a $T_0$-space, $G$ is regular (since $\{e\}$ is closed).*

**From now on, we always assume that our topological group $G$ is Hausdorff**.

**Definition 1.17.** *Define $G^0$ to be the connected component of $G$ containing $e$. It is usually called the* ***identity component*** *of $G$.*

The following is a useful lemma.

**Lemma 1.18** ([Sep07, Theorem 1.15])**.** *Let $U$ be a connected open neighbourhood of $e$. Then $G^0 = \bigcup_{n=1}^{\infty} U^n$. Thus, $G^0$ is both open and closed in $G$.*

*Proof.* Take an open symmetric neighbourhood $V = U \cap U^{-1}$ contained in $U$ (modify the proof of Lemma 1.13, then you get it). Set $H = \bigcup_{n=1}^{\infty} V^n$. By construction, $H$ is a connected open subgroup of $G$ containing $e$. Hence it is also a connected open subgroup of $G^0$. Then $G^0$ is a disjoint union of open cosets $x_jH$ for $x_j$. Connectedness of $G^0$ implies that there is only one such coset. Hence $G^0 = H$. As connected component is always closed, $G^0$ is closed. □

**Theorem 1.19.** *$G^0$ is a closed normal subgroup of $G$. Hence $\pi_0(G)$ inherits a group structure $\pi_0(G) \simeq G/G^0$.*

*Proof.* The fact that $G^0$ is a closed subgroup is proved in Lemma 1.18. We only need to prove $G^0$ is normal. Let $g \in G$, then $gG^0g^{-1}$ is isomorphic to $G^0$ as a topological group. As both $G^0$ and $gG^0g^{-1}$ contain $e$, they must be equal. Therefore, $G^0$ is normal. The last statement is now trivial. □

## 2. Haar Measure and Modular Function

From my understanding, measure theory is developed to give a foundation of integration, including integrable functions and admissible domains (i.e. measurable subsets). Explicit measure theory is both axiomatic and constructive. For instance, in Lebesgue measure theory, we start from measurable sets, then move on to measurable functions and integrable functions. Yet in practice, at least in representation theory, we rarely deal with *bad* functions and *bad* integration domains. Motivated from this, we develop Haar measure from a functional-analytic point of view. For a quick summary on topology of certain function spaces such as $C_c(G)$, see the appendix.

By a **locally compact group**, we mean a Hausdorff topological group that is locally compact as a topological space. Let $G$ be a locally compact group. As usual, $C_c(G)$ is the complex vector space of $\mathbb{C}$-valued functions on $G$ that are compactly supported. Equip $C_c(G)$ with the canonical $LF$-space topology.

**Remark 2.1.** *A more serious course on representation theory should now discuss the representation of $G$ on several standard function spaces, such as $C_c(G)$. Yet this requires students to be very familiar with functional analysis. As this is a first course on Lie theory, students are suggested to take facts for granted. For instance, we will shortly use the fact that the right translation of $G$ on $C_c(G)$ is continuous.*

**Definition 2.2.** *A measure $\mu$ on a locally compact group $G$ is said to be* ***left (resp. right) invariant*** *if $d\mu(gx) = d\mu(x)$ ($d\mu(xg) = d\mu(x)$) for all $x, g \in G$. To be precise, for all $f \in C_c(G)$ and $g \in G$,*

$$\int_G f(x)d\mu(x) = \int_G f(g^{-1}x)d\mu(x) \quad (\textit{resp.} \int_G f(x)d\mu(x) = \int_G f(xg^{-1})d\mu(x)).$$

**Theorem 2.3** ([Rei-Ste00, Theorem 3.3.2])**.** *On a locally compact group, there exists a left (resp. right) invariant positive measure, called the* ***left*** *(resp.* ***right****)* ***Haar measure****, not identically zero and uniquely determined up to a constant factor.*

As our treatment is non-constructive, we will not be able to prove the above theorem. In fact, Theorem 2.3 is far more important than its proof.

**Remark 2.4.** *An explicit formula of a (left or right) Haar measure of a given group is difficult, especially for compact groups. In practice, one needs to use certain decomposition of Haar measures to reduce the Haar measure of a big group into product of measures of small groups. I do not hope we can reach that far, but for interested students, see* [Kna02, Chapter VIII].

**Definition 2.5.** *Usually we write $d_l g$ (resp. $d_r g$) for a left (resp. right) Haar measure on a locally compact group $G$.*

**Example 2.6.** *For $G = \mathbb{R}$, the $d_l g = d_r g$ can be chosen to be the Lebesgue measure on $\mathbb{R}$:*

$$\int_{\mathbb{R}} f(x+a)\,dx = \int_{\mathbb{R}} f(a+x)\,dx = \int_{\mathbb{R}} f(x)\,dx.$$

**Remark 2.7.** *One should be careful of the potentially confusing notation here. One usually write $dx$ both as a differential form and as a measure. As a differential form, $d(-x) = -dx$;*

*as a measure $d(-x) = dx$. This does not affect integration since we need to fix an orientation when integrating a top differential form, i.e.*

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx.$$

*In general, under a change of variables, differential forms change by Jacobian determinant; while measures change by absolute value of Jacobian determinant.*

**Example 2.8.** *For $G = \mathbb{R}^\times$, the $d_l x = d_r x$ can be chosen to be $\frac{dx}{|x|}$:*

$$\frac{d(ax)}{|ax|} = \frac{dx}{|x|}.$$

**Example 2.9.** (1) *For $G = \mathrm{GL}_n(\mathbb{R})$, we write $g = (g_{ij})$, then the left Haar measure (and later we will see that it is also the right Haar measure) is*

$$dg = \frac{1}{|\det g|^n} dg_{ij}.$$

(2) *For $G = \mathrm{GL}_n(\mathbb{C})$, we write $g = (g_{ij})$, then the left Haar measure (and later we will see that it is also the right Haar measure) is*

$$dg = \frac{1}{|\det g|^{2n}} dg_{ij} = \frac{1}{|\det g|_{\mathbb{C}}^n} dg_{ij}.$$

*For a detailed proof of the Haar measure of $\mathrm{GL}_n(\mathbb{R})$ given above, see* [Gold06, Proposition 1.4.3].

**Proposition 2.10.** *The left Haar measure of $G$ is finite (i.e. $\int_G 1 d_l g$ is finite) if and only if $G$ is compact.*

*Proof.* If $G$ is compact, then the constant function 1 belongs to $C_c(G) = C(G)$. Hence $\int_G 1\, d_l g$ is finite.

Conversely if $G$ is noncompact, we need to show that $\int_G 1 d_l g$ is infinite. (One may think this is trivial as $1 \notin C_c(G)$ when $G$ is noncompact. Yet this is not precise. What we need to prove in nature is that there is no continuous linear extension of the linear functional $f \mapsto \int_G f d_l g$ to a larger space containing the constant function 1.) Take an arbitrary nonzero function $f \in C_c(G)$ with $0 < f(x) \leq 1$ for all $x \in G$ and $\int_G f(g) d_l g \neq 0$. Set $K = \mathrm{supp}(f)$ to be a compact subset. As $G$ is non-compact, we are able to find a sequence of mutually disjoint compact subsets $K_j = a_j K$ of $G$. Then for any positive integer $n$,

$$\int_G 1\, d_l g \geq \sum_{j=1}^n \int_{K_j} 1\, d_l g \geq \sum_{j=1}^n \int_{K_j} (L_{a_j^{-1}} f)(g)\, d_l g = n \int_K f(g)\, d_l g.$$

□

A factor, which turns out to be a character called **modular character**, links the left and right Haar measures.

**Definition 2.11.** *Let $d_l g$ be a left Haar measure on a topological group $G$. Then we define a function on $G$ such that*

$$d_l(xy) = \delta_G(y)^{-1} d_l x.$$

*This function is called the* ***modular function****.*

**Remark 2.12.** *The notation of modular characters are quite confusing. Here we follow N. Wallach's notation. In* [Kna02]*, Knapp's* $\Delta_G$ *is our* $\delta_G$*; in* [GWar72]*, Warner's* $\delta_G$ *is our* $\delta_G^{-1}$*. In other references or online notes, other authors (such as F. Murnaghan, see her online notes on representation theory of p-adic reductive groups) also define* $\Delta_G$ *to be* $\delta_G^{-1}$*.*

**Proposition 2.13** ([Rei-Ste00, Page 90])**.** *The modular function* $\delta_G$ *is a continuous character which takes values in* $\mathbb{R}_+^\times$*.*

*Sketch of proof.* It is obvious that $\delta_G$ takes value in $\mathbb{R}_+^\times$ ($\delta_G(y)^{-1}$ is the absolute value of a Jacobian determinant) and $\delta_G \to \mathbb{R}_+^\times$ is a group homomorphism. Only continuity is an issue.

Pick $f \in C_c(G)$ with $\int_G f(x)d_lx \neq 0$. Then

$$\delta_G(y)^{-1}\int_G f(x)d_lx = \int_G f(x)d_l(xy) = \int_G f(xy^{-1})\,d_lx.$$

Thus,

$$\delta_G(y) = \frac{\int_G (R_yf)(x)d_lx}{\int_G f(x)d_lx}.$$

Then the proposition follows from the fact that the right translation of $G$ on $C_c(G)$ is continuous. (In other words, $C_c(G)$ is a continuous representation of $G$. For the definition of continuous representation on a locally convex topological vector space and some related examples, see [GWar72, Section 4.1].) □

**Corollary 2.14** ([Kna02, Corollary 8.30])**.** *We write* $d_rg$ *for the right Haar measure of* $G$*.*

(1) $d_l(x^{-1}) = \delta_G(x)d_lx$ *are right Haar measures.*
(2) $d_r(x^{-1}) = \delta_G(x)^{-1}d_rx$ *are left Haar measures.*
(3) $d_r(yx) = \delta_G(y)d_rx$

*Proof.* We only prove 1), the other two can be checked in the same way. By a change of variable, we have

$$\int_G f(xy)\delta_G(x)d_lx = \int_G f(x)\delta_G(xy^{-1})d_l(xy^{-1}) = \int_G f(x)\delta_G(x)d_lx.$$

Thus, $\delta_G(x)d_lx$ is a right Haar measure. It must be proportional to $d_l(x^{-1})$. Set $c > 0$ to be a constant such that $d_l(x^{-1}) = c\delta_G(x)d_lx$. Then by changing $x \mapsto x^{-1}$, we get $c^2d_lx = d_lx$, hence $c = 1$. □

**Example 2.15.** *Let*

$$G := \left\{\begin{pmatrix} x & y \\ & 1\end{pmatrix} \in \mathrm{GL}_2(\mathbb{R})\right\}.$$

*This is a semidirect product of* $\mathbb{R}^\times$ *and* $\mathbb{R}$*. Compute*

$$\begin{pmatrix} a & b \\ & 1\end{pmatrix}\cdot\begin{pmatrix} x & y \\ & 1\end{pmatrix} = \begin{pmatrix} ax & ay+b \\ & 1\end{pmatrix}.$$

*So the left Haar measure* $d_lg = |x|^{-1}d^\times xdy$*. Compute*

$$\begin{pmatrix} x & y \\ & 1\end{pmatrix}\cdot\begin{pmatrix} a & b \\ & 1\end{pmatrix} = \begin{pmatrix} ax & y+bx \\ & 1\end{pmatrix}.$$

*Hence the right Haar measure* $d_rg = d^\times xdy$*. As* $d^\times(ax)d(ay+b) = |a|d^\times xdy$*,*

$$\delta_G(\begin{pmatrix} a & b \\ & 1\end{pmatrix}) = |a|.$$

**Definition 2.16.** *We say a topological group $G$ is **unimodular** if $\delta_G = 1$. In this case, the left Haar measure and right Haar measure coincide.*

**Corollary 2.17.** *Clearly, compact groups and locally compact abelian groups are unimodular.*

*Proof.* Obvious.

□

## 3. Lie Groups, Homomorphisms and Smoothness

**Definition 3.1.** *A **Lie group** $G$ is a smooth manifold (which is always assumed to be second countable and Hausdorff) and a group on which the multiplication and inverse operation of $G$ is smooth.*

Though the definition seems quite natural and simple, there are some of hidden issues to be clarified. As this course focuses on heavily on structure theory and representation theory, hence quite a lot of foundational work will be taken for granted. Explanations will be given, but not proofs. Honestly, I am not an expert in differential geometry/topology, and I never have the motivation and opportunity to *at least* read proofs of some foundational theorems.

**Example 3.2.** *Obviously, $\mathbb{R}^n$ (especially for $\mathbb{R}^4$), equipped with the standard Euclidean smooth structure and the usual abelian group structure under addition, is a Lie group.*

**Example 3.3.** *As the group multiplication and inverse are rational functions on $\mathrm{GL}_n(\mathbb{R})$ and $\mathrm{GL}_n(\mathbb{C})$, $\mathrm{GL}_n(\mathbb{R})$ and $\mathrm{GL}_n(\mathbb{C})$ are Lie groups.*

Let $M$ be a smooth manifold, recall that a smooth manifold $N$ is said to be a **submanifold** of $M$ if there exists an injective **immersion** $\iota : N \hookrightarrow M$. Here by an **immersion**, we mean a smooth map $\varphi$ whose differential $d\varphi$ is non-singular (meaning that $\ker d\varphi = 0$). A more useful notion is **embedding**. Recall that $N$ embeds into $M$ if there exists an injective immersion $\varphi : N \hookrightarrow M$ such that $\varphi$ is a homeomorphism onto its image (i.e. the relative topology of $\varphi(N)$ coincides with the topology transferred from $N$). Usually, if $N$ is a subset of $M$ and $\iota$ is the inclusion map, we often omit $\iota$ and just say $N$ is a submanifold of $M$.

Let $G$ be a Lie group.

**Definition 3.4.** *A **Lie subgroup** of $G$ consists of the following data: a Lie group $H$ and an injective immersion $\varphi : H \hookrightarrow G$ such that $\varphi$ is an abstract group homomorphism. If $H$ is just a subset of $G$ and $\varphi$ is the inclusion map, then we often omit $\varphi$ and just say $H$ is a Lie subgroup of $G$. We say $H$ is a **closed subgroup** of $G$ if $\varphi(H)$ is a closed subset of $G$.*

**Definition 3.5.** *A **homomorphism of Lie groups** $\varphi : G \to H$ is a homomorphism of abstract groups that is smooth. We say $\varphi : G \to H$ is an **isomorphism** of Lie groups if $\varphi$ is a homomorphism of Lie groups that is also a diffeomorphism of smooth manifolds.*

The following theorem is quite useful to identify Lie subgroups.

**Theorem 3.6** ([FWar83, Theorem 3.21, 3.42])**.**
(1) *Let $G$ be a Lie group. Assume that $H$ is an abstract subgroup that is also a closed subset of $G$, then under the relative topology, $H$ is a Lie subgroup of $G$.*
(2) *Suppose $(H, \varphi)$ is a Lie subgroup of $G$, then $\varphi$ is an embedding if and only if $\varphi(H)$ is closed in $G$.*

**Example 3.7.** *Equip $\mathbb{C}^\times$ the with the usual Euclidean topology (inherited from $\mathbb{C}$) and the usual group structure under multiplication of complex numbers. Then $\mathbb{C}^\times$ is a Lie group. The one-dimensional sphere $S^1 = \{z \in \mathbb{C} \,|\, |z| = 1\}$ is a closed subgroup of $\mathbb{C}^\times$. Theorem 3.6 says that once equipped $S^1$ with the relative topology, it is a Lie subgroup of $\mathbb{C}^\times$.*

Example 3.8 may intrigue students studying dynamics; for students in representation theory, we are more into groups that are better-behaved (Example 3.9).

**Example 3.8.** *Let $a \in \mathbb{R}$ be an irrational number. Consider the map*

$$\varphi : \mathbb{R} \to S^1 \times S^1, t \mapsto (e^{2\pi \mathbf{i} t}, e^{2a\pi \mathbf{i} t}).$$

*Obviously $\varphi$ is an injective Lie group homomorphism. Thus, we can realized $\mathbb{R}$ as a Lie subgroup of $S^1 \times S^1$ via $\varphi$. But $\varphi(\mathbb{R})$ is not closed! In fact, the relative topology on $\varphi(\mathbb{R})$ is not the same as the topology on $\mathbb{R}$.*

**Example 3.9.** *Let $F = \mathbb{R}$ or $\mathbb{C}$. If an abstract group $G \leq \mathrm{GL}_m(F)$ is defined by polynomial equations, then $G$ is a closed subset of $\mathrm{GL}_m(F)$ (preimage of $\{0\}$ is closed, intersection of closed subsets is closed). Hence under the relative topology of $\mathrm{GL}_m(F)$, $G$ is a Lie subgroup. This implies that the following are all closed (Lie) subgroups of $\mathrm{GL}_m(F)$*

$$\mathrm{SL}_m(F) := \{g \in \mathrm{GL}_m(F) \,|\, \det g = 1\};$$

$$\mathrm{Sp}_{2n}(F) := \left\{ g \in \mathrm{GL}_{2n}(F) \,\middle|\, g \begin{pmatrix} & J_n \\ -J_n & \end{pmatrix} g^t = \begin{pmatrix} & J_n \\ -J_n & \end{pmatrix} \right\}.$$

*The complex orthogonal groups are Lie subgroups of $\mathrm{GL}_m(\mathbb{C})$ (we only take care of smooth structure at the moment):*

$$\mathrm{O}_n(\mathbb{C}) := \left\{ g \in \mathrm{GL}_n(\mathbb{C}) \,\middle|\, g J_n g^t = J_n \right\};$$

$$\mathrm{SO}_n(\mathbb{C}) := \left\{ g \in \mathrm{O}_n(\mathbb{C}) \,\middle|\, \det g = 1 \right\}.$$

*The real orthogonal groups (used in geometry) are Lie subgroups of $\mathrm{GL}_{p+q}(\mathbb{R})$:*

$$\mathrm{O}(p,q) := \left\{ g \in \mathrm{GL}_{p+q}(\mathbb{R}) \,\middle|\, g \begin{pmatrix} \mathrm{I}_p & \\ & -\mathrm{I}_q \end{pmatrix} g^t = \begin{pmatrix} \mathrm{I}_p & \\ & -\mathrm{I}_q \end{pmatrix} \right\};$$

$$\mathrm{SO}(p,q) := \left\{ g \in \mathrm{O}(p,q) \,\middle|\, \det g = 1 \right\}.$$

*The unitary groups (used in geometry) are Lie subgroups of $\mathrm{GL}_{p+q}(\mathbb{C})$:*

$$\mathrm{U}(p,q) := \left\{ g \in \mathrm{GL}_{p+q}(\mathbb{C}) \,\middle|\, g \begin{pmatrix} \mathrm{I}_p & \\ & -\mathrm{I}_q \end{pmatrix} \overline{g}^t = \begin{pmatrix} \mathrm{I}_p & \\ & -\mathrm{I}_q \end{pmatrix} \right\};$$

$$\mathrm{SU}(p,q) := \{g \in \mathrm{U}(p,q) \,|\, \det g = 1\}.$$

**Example 3.10.** *Let $F = \mathbb{R}$ or $\mathbb{C}$. Consider the group*

$$G := \left\{ g \in \mathrm{GL}_{2n}(F) \,\middle|\, g \begin{pmatrix} & \mathrm{I}_n \\ -\mathrm{I}_n & \end{pmatrix} g^t = \begin{pmatrix} & \mathrm{I}_n \\ -\mathrm{I}_n & \end{pmatrix} \right\}.$$

*For every $g \in G$, we claim that $\begin{pmatrix} \mathrm{I}_n & \\ & J_n \end{pmatrix} g \begin{pmatrix} \mathrm{I}_n & \\ & J_n \end{pmatrix} \in \mathrm{Sp}_{2n}(F)$. Indeed,*

$$\begin{aligned}
&\begin{pmatrix} \mathrm{I}_n & \\ & J_n \end{pmatrix} g \begin{pmatrix} \mathrm{I}_n & \\ & J_n \end{pmatrix} \begin{pmatrix} & \mathrm{I}_n \\ -\mathrm{I}_n & \end{pmatrix} \begin{pmatrix} \mathrm{I}_n & \\ & J_n \end{pmatrix} g^t \begin{pmatrix} \mathrm{I}_n & \\ & J_n \end{pmatrix} \\
&= \begin{pmatrix} \mathrm{I}_n & \\ & J_n \end{pmatrix} g \begin{pmatrix} & J_n \\ -J_n & \end{pmatrix} g^t \begin{pmatrix} \mathrm{I}_n & \\ & J_n \end{pmatrix} \\
&= \begin{pmatrix} \mathrm{I}_n & \\ & J_n \end{pmatrix} \begin{pmatrix} & J_n \\ -J_n & \end{pmatrix} \begin{pmatrix} \mathrm{I}_n & \\ & J_n \end{pmatrix} \\
&= \begin{pmatrix} & \mathrm{I}_n \\ -\mathrm{I}_n & \end{pmatrix}.
\end{aligned}$$

*Thus, $G \simeq \mathrm{Sp}_{2n}(F)$, $\quad g \mapsto \begin{pmatrix} \mathrm{I}_n & \\ & J_n \end{pmatrix} g \begin{pmatrix} \mathrm{I}_n & \\ & J_n \end{pmatrix}$. The skew-symmetric bilinear form determined by $\begin{pmatrix} & \mathrm{I}_n \\ -\mathrm{I}_n & \end{pmatrix}$ is often used in geometry or earlier work of automorphic forms and representation theory (such as Siegel modular forms); while the form determined by $\begin{pmatrix} & J_n \\ -J_n & \end{pmatrix}$ is more commonly used in automorphic forms and representation theory in recent years.*

**Example 3.11.** *The group $\mathrm{SL}_2(\mathbb{R}) = \mathrm{Sp}_2(\mathbb{R})$ can be embedded into $\mathrm{Sp}_4(\mathbb{R})$ via*

$$g \mapsto \begin{pmatrix} 1 & & \\ & g & \\ & & 1 \end{pmatrix}.$$

*The group $\mathrm{GL}_2(\mathbb{R})$ can be embedded into $\mathrm{Sp}_4(\mathbb{R})$ via*

$$g \mapsto \begin{pmatrix} g & \\ & J_2 (g^{-1})^t J_2 \end{pmatrix}.$$

*This subgroup is usually called the standard Siegel Levi subgroup of $\mathrm{Sp}_4(\mathbb{R})$. Then $\mathrm{SL}_2(\mathbb{R})$ can also embedded into $\mathrm{Sp}_4(\mathbb{R})$ via*

$$\mathrm{SL}_2(\mathbb{R}) \hookrightarrow \mathrm{GL}_2(\mathbb{R}) \hookrightarrow \mathrm{Sp}_4(\mathbb{R}).$$

Note that when we say $G$ is a Lie group, we implicitly include a smooth structure on $G$. Then it is naturally to ask the existence and uniqueness of such smooth structure (Hilbert's 5-th problem). We summarize them in the following two theorems. The following is a *weak* version of Gleason-Montgomery-Zippin-Yamabe's work on Hilbert's 5-th problem, yet it is sufficient for our use.

**Theorem 3.12** (Gleason-Montgomery-Zippin-Yamabe)**.** *Every second countable locally Euclidean topological group $G$ can be equipped with a unique smooth structure so that $G$ is a Lie group.*

**Remark 3.13.** *I am not an expert in history. I guess the uniqueness may be known at the time of Hilbert (as it is not very difficult and is usually covered in standard textbook nowadays, see the next theorem). Only the existence is critical, which is confirmed by Gleason, Montgomery, Zippin and Yamabe.*

The uniqueness of the smooth structure on a second countable locally Euclidean topological group is in fact guaranteed by the following theorem.

**Theorem 3.14** ([FWar83, Theorem 3.39])**.** *A continuous homomorphism between Lie groups is automatically smooth. As a consequence, if there are two different Lie group structures on the same second countable locally Euclidean topological group $G$, then the identity map provides an isomorphism of Lie groups.*

**Example 3.15.** *There are uncountably many smooth structures on $\mathbb{R}^4$ (due to the work of C. Taubes (1987)). Gleason-Montgomery-Zippin-Yamabe's theorem says that there is only one (the usual euclidean smooth structure) to make it a Lie group.*

## 4. Lie Algebras of Lie Groups and Correspondence Theorem

Recall that a smooth vector field on a smooth manifold $M$ is an assignment: to each $p \in M$, assign a tangent vector $X_p \in T_pM$ such that this assignment depends on $p$ smoothly. (This is not a definition! A rigorous definition should be the following: *a smooth vector field is defined to be a smooth section of the tangent bundle.* We are just trying to minimize our terminology here.) For two smooth vector fields, one can define their Lie bracket $[X, Y]$:

$$[X, Y]_p(f) := X_p(Yf) - Y_p(Xf).$$

It is well-known that if $X, Y$ are smooth, so is $[X, Y]$. Moreover, the Lie bracket of smooth vector fields is skew symmetric and satisfies Jacobi identity:

$$[[X, Y], Z] + [[Y, Z], X] + [[Z, X], Y] = 0$$

as smooth vector fields. The space of all smooth vector fields on $M$, denoted by $\Gamma(TM)$, forms an infinite-dimensional Lie algebra over $\mathbb{R}$. In the following, when $M, N$ are smooth manifolds, we write $C^\infty(M, N)$ for the space of smooth maps from $M$ to $N$. In particular, when $N = \mathbb{C}$, we just write $C^\infty(M)$ for simplicity.

**Definition 4.1.** *Let $\varphi : M \to N$ be a smooth map between smooth manifolds $M, N$. Let $X \in \Gamma(TM)$, $Y \in \Gamma(TN)$. We say $X$ and $Y$ are $\varphi$-related if for any $p \in M$, $d\varphi(X_p) = Y_{\varphi(p)}$.*

In the following, a key lemma from differential geometry plays an essential role.

**Lemma 4.2** ([FWar83, Proposition 1.55])**.** *Let $\varphi \in C^\infty(M, N)$, $X_1, X_2 \in \Gamma(TM)$, $Y_1, Y_2 \in \Gamma(TN)$. If $X_j$ is $\varphi$-related to $Y_j$ ($j = 1, 2$), then $[X_1, X_2]$ is $\varphi$-related to $[Y_1, Y_2]$.*

Let $G$ be a Lie group. Consider the left (resp. right) translation on $G$:

$$L_g(x) := gx \qquad (\text{resp. } R_g(x) = xg).$$

A vector field $X$ on $G$ is said to be left-(resp. right-)invariant if $dL_gX = X$ (resp. $dR_gX = X$), i.e. $dL_g(X_p) = X_{gp}$ (resp. $dR_g(X_p) = X_{pg}$). In other words, a left-(resp. right-)invariant vector field is related to itself by any left (resp. right) translation. A left-(resp. right-)invariant vector field is completely determined by its value at identity. This further implies that the space of left-(resp. right-)invariant vector field is finite-dimensional (which is equal to $\dim G = \dim T_eG$).

**Proposition 4.3** ([FWar83, Proposition 3.7]). (1) *Every left-(resp. right-)invariant vector field $X$ is smooth.*
(2) *The Lie bracket of two smooth left-(resp. right-)invariant vector fields is also left-(resp. right-)invariant.*

Thus, the space of all left-(resp. right-)invariant invariant vector field forms a finite-dimensional Lie subalgebra of $\Gamma(TM)$. This underlying vector space of this Lie algebra is identified with the tangent space $T_eG$ via the evaluation map $X \mapsto X_e$. The only issue is that how the Lie brackets defined from left-invariant vector fields and the right-invariant vector fields are related.

Write $\Gamma_l(TM)$ (resp. $\Gamma_r(TM)$) for the Lie algebra of left-(resp. right-)invariant vector fields. Consider the diffeomorphism $I : G \to G, \quad g \mapsto g^{-1}$. Then $R_g \circ I = I \circ L_{g^{-1}}$. This implies that the differential $dI$ maps left invariant vector fields to right-invariant vector fields and vice-versa. Thus the map $-dI$ is an anti-isomorphism of Lie algebras $\Gamma_l(TM) \simeq \Gamma_r(TM)$. The differential $-dI|_e$ is exactly the identity map on $T_eG$. Thus, the Lie algebra structures on $T_eG$ inherited from left and right invariant vector fields are just related by an anti-isomorphism. So, we gain nothing new from considering right-invariant vector fields.

In view of the above discussion, unless we state otherwise, we only consider the Lie algebra structure inherited from left-invariant vector fields.

**Definition 4.4.** *We will call the space of left-invariant vector fields on a Lie group $G$* ***the Lie algebra of*** *$G$, usually denoted by the German letter $\mathfrak{g}$.*

**Example 4.5.** *Let $G = \mathbb{R}$. Define $X \in \Gamma(TM)$, $X_p(f) := \frac{d}{dx}\big|_{x=p} f(x)$. Then $X$ is a left-invariant vector field on $\mathbb{R}$. This is obvious from the identity*

$$X_p(f) = \frac{d}{dx}\Big|_{x=p} f(x) = \frac{d}{dx}\Big|_{x=0} f(x+p) = (dL_p X_0)(f).$$

*This means that the Lie algebra $\mathfrak{g} = \mathbb{R} \cdot \frac{d}{dr}$ (when we realize Lie algebra as space of invariant vector fields).*

**Example 4.6.** *Let us show that the Lie algebra of the Lie group $\mathrm{GL}_n(\mathbb{R})$ defined by left-invariant vector field is exactly $\mathfrak{gl}_n(\mathbb{R})$, where the Lie bracket is the usual one, i.e. $[A, B] = AB - BA$. To distinguish to two definitions, we write $X^A$ for the left-invariant vector field on $\mathrm{GL}_n(\mathbb{R})$ determined by $A \in \mathfrak{gl}_n(\mathbb{R}) \simeq T_{\mathrm{I}}(\mathrm{GL}_n(\mathbb{R}))$. We aim to prove that*

$$X^{[A,B]} = [X^A, X^B].$$

*The elementary matrix $E_{ij}$ form a basis of $\mathrm{Mat}_n(\mathbb{R})$. By linearity, we only need to check the above formula for $E_{ij}$. Let $(x_{ij})$ be the standard coordinate on $\mathrm{Mat}_n(\mathbb{R})$. We compute the vector field corresponding to $E_{ij}$. Take $g \in \mathrm{GL}_n(\mathbb{R})$, since for small $t$, $\mathrm{I}_n + tE_{ij}$ is invertible, we have*

$$X_g^{E_{ij}} f = \frac{d}{dt}\Big|_{t=0} f(g(\mathrm{I} + tE_{ij})) = \sum_{k=1}^{n} x_{ki}(g) \frac{\partial f}{\partial x_{kj}}\Big|_g.$$

*Thus, $X^{E_{ij}}$ is the vector field*

$$X^{E_{ij}} = \sum_{k=1}^{n} x_{ki} \frac{\partial}{\partial x_{kj}}.$$

*Then it is just a routine work to check*

$$[X^{E_{ij}}, X^{E_{kl}}] = X^{[E_{ij}, E_{kl}]}.$$

*The formula for Lie bracket of vector fields is well-known: if* $X = \sum f_i \frac{\partial}{\partial x_i}, Y = \sum g_j \frac{\partial}{\partial x_j}$, *then*

$$[X, Y] = \sum_i \sum_j (f_j \frac{\partial g_i}{\partial x_j} - g_j \frac{\partial f_i}{\partial x_j}) \frac{\partial}{\partial x_i}.$$

**Proposition 4.7** ([FWar83, Theorem 3.14, 3.16])**.** (1) *Suppose* $\varphi : G \to H$ *is a Lie group homomorphism, for every* $X \in \Gamma_l(TM)$, *define* $d\varphi(X)$ *to be the unique left-invariant vector field on* $H$ *whose value at the identity is* $d\varphi(X_e)$. *Then* $d\varphi : \mathfrak{g} \to \mathfrak{h}$ *is a Lie algebra homomorphism.*

(2) *If we further assume that* $G$ *is connected, then* $\varphi$ *is uniquely determined by* $d\varphi$. *In other words, if we have two homomorphisms* $\varphi_1$ *and* $\varphi_2$ *with the property that* $d\varphi_1 = d\varphi_2$, *then* $\varphi_1 = \varphi_2$.

*Proof.* **Proof of (1)**: Let $X, Y \in \Gamma_l(TM)$. We need to prove that

$$d\varphi([X, Y]) = [d\varphi(X), d\varphi(Y)].$$

By definition, we only need to prove that the left-hand side and the right-hand side agrees at identity. Define $\widetilde{X} := d\varphi(X)$. Then one can easily check by definition that

$$\widetilde{X}_{\varphi(g)} = d\varphi(X_g) \qquad \forall g \in G.$$

This implies that $X$ and $d\varphi(X)$ are $\varphi$-related. Similarly, $Y$ and $d\varphi(Y)$ are $\varphi$-related. Lemma 4.2 implies that $[X, Y]$ and $[d\varphi(X), d\varphi(Y)]$ are $\varphi$-related, i.e.

$$[d\varphi(X), d\varphi(Y)]_{\varphi(g)} = d\varphi([X, Y]_g).$$

In particular, for $g = e$, we have

$$[d\varphi(X), d\varphi(Y)]_e = d\varphi([X, Y]_e).$$

This is exactly what we need to prove.

**Proof of (2)**: The proof requires the theory of differential ideals on a smooth manifold, which is beyond the capacity of this course. We omit it. □

**Theorem 4.8** ([FWar83, Theorem 3.19])**.** *There is a one-to-one correspondence between connected (not necessarily closed) Lie subgroups of* $G$ *and Lie subalgebras of* $\mathfrak{g}$.

*Proof.* In fact, if $\mathfrak{h}$ is a Lie subalgebra of $\mathfrak{g}$, then the maximal connected integral submanifold through $e$ of the involutive distribution determined by $\mathfrak{h}$ admits a Lie subgroup structure. The details are beyond the capacity of this notes, hence will be omitted. □

## 5. Compactness of U($n$), O($n$) and Sp($2n$)

The **Hamiltonian quaternion** $\mathbb{H}$ is a four dimensional (noncommutative but associative) division algebra over $\mathbb{R}$. As a vector space, it has a basis $1, \mathbf{i}, \mathbf{j}, \mathbf{k}$ satisfying the multiplication rules:

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -1, \quad \mathbf{ij} = -\mathbf{ji} = \mathbf{k}, \quad \mathbf{jk} = -\mathbf{kj} = \mathbf{i}, \quad \mathbf{ki} = -\mathbf{ik} = j.$$

The center of H is $\mathbb{R}$, identified with $\mathbb{R} \cdot 1 \subseteq \mathbb{H}$. The **conjugation** in $\mathbb{H}$ is defined via:

$$\overline{a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}} = a - b\mathbf{i} - c\mathbf{j} - d\mathbf{k}.$$

The topology of $\mathbb{H}$ is defined via the identification $\mathbb{H} \simeq \mathbb{R}^4$ as a vector space, for instance, its topology can be defined via the norm

$$|x| := \sqrt{x \cdot \overline{x}}.$$

The set $\mathbb{H}^\times$ forms a non-abelian Lie group, usually denoted by $\mathrm{GL}_1(\mathbb{H})$. Theory of $\mathrm{GL}_n(\mathbb{H})$ can be developed in a similar way as $\mathrm{GL}_n(\mathbb{R})$ and $\mathrm{GL}_n(\mathbb{C})$, if we are careful of our wording as $\mathbb{H}$ is non-abelian.

Regard $\mathbb{H}^n$ as a *right* $\mathbb{H}$-module. Realize every element in $\mathbb{H}^n$ as a column vector. Elements in $\mathrm{Mat}_n(\mathbb{H})$ acts on $\mathbb{H}^n$ on the *left* by matrix multiplication. As left multiplication of matrix commutes with right scalar multiplication, $\mathrm{Mat}_n(\mathbb{H})$ is identified with the set of right $\mathbb{H}$-module homomorphisms. Thus, we define

$$\mathrm{GL}_n(\mathbb{H}) := \{f : \mathbb{H}^n \to \mathbb{H}^n \mid f \text{ is a right } \mathbb{H}\text{-module isomorphism}\}.$$

As multiplication in $\mathbb{H}$ is not commutative, hence it is not obvious that $\mathrm{GL}_n(\mathbb{H})$ is an open submanifold of $\mathrm{Mat}_n(\mathbb{H})$. Maybe a modification of det still works, yet a more convenient way to describe $\mathrm{GL}_n(\mathbb{R})$ is as follows.

Every element in $\mathbb{H}$ can be uniquely written as $a+\mathbf{j}b$ with $a, b \in \mathbb{C}$. Hence we can identify

$$\mathbb{H}^n \simeq \mathbb{C}^{2n}, \quad (a_1 + \mathbf{j}b_1, \cdots, a_n + \mathbf{j}b_n) \mapsto (a_1, a_2, \cdots, a_n, b_1, \cdots, b_n).$$

as right $\mathbb{C}$-modules. Also, each matrix in $\mathrm{Mat}_n(\mathbb{H})$ can be uniquely written as $A + \mathbf{j}B$ with $A, B \in \mathrm{Mat}_n(\mathbb{C})$.

**Lemma 5.1.** *$\iota : \mathrm{Mat}_n(\mathbb{H}) \to \mathrm{Mat}_{2n}(\mathbb{C}), \quad A + \mathbf{j}B \mapsto \begin{pmatrix} A & -\overline{B} \\ B & \overline{A} \end{pmatrix}$ is an injective homomorphisms of $\mathbb{R}$-algebras.*

*Proof.* Injectivity is obvious. Note that for $a+b\mathbf{i} \in \mathbb{C}$, $(a+b\mathbf{i})\mathbf{j} = a\mathbf{j}+b\mathbf{i}\mathbf{j} = \mathbf{j}(a-b\mathbf{i}) = \mathbf{j}\cdot\overline{a+b\mathbf{i}}$. Thus for any $A \in \mathrm{Mat}_n(\mathbb{C})$, $A\mathbf{j} = \mathbf{j}\overline{A}$. Then it is easy to check

$$\iota(A + \mathbf{j}B) \cdot \iota(C + \mathbf{j}D) = \iota((A + \mathbf{j}B) \cdot (C + \mathbf{j}D)).$$

□

By Lemma 5.1, $\mathrm{Mat}_n(\mathbb{H})$ is a closed subspace of $\mathrm{Mat}_{2n}(\mathbb{C})$. Obviously, $X \in \mathrm{Mat}_n(\mathbb{H})$ is invertible if and only if $\iota(X)$ is invertible. Thus, we identify $\mathrm{GL}_n(\mathbb{H})$ as a closed subgroup of $\mathrm{GL}_{2n}(\mathbb{C})$ as follows.

**Corollary 5.2.** *$\mathrm{GL}_n(\mathbb{H}) \overset{\iota}{\simeq} \left\{ g = \begin{pmatrix} A & -\overline{B} \\ B & \overline{A} \end{pmatrix} \in \mathrm{Mat}_{2n}(\mathbb{C}) \,\middle|\, \det g \neq 0 \right\}$ is a closed Lie subgroup of $\mathrm{GL}_{2n}(\mathbb{C})$.*

**Remark 5.3.** *It is easy to check that*

$$\iota(\mathrm{Mat}_n(\mathbb{H})) = \left\{ g \in \mathrm{Mat}_{2n}(\mathbb{C}) \,\middle|\, g \begin{pmatrix} & -\mathrm{I}_n \\ \mathrm{I}_n & \end{pmatrix} = \begin{pmatrix} & -\mathrm{I}_n \\ \mathrm{I}_n & \end{pmatrix} \overline{g} \right\}.$$

**Definition 5.4.** *Define the* ***compact symplectic group***

$$\mathrm{Sp}(2n) := \{g \in \mathrm{GL}_n(\mathbb{H}) \mid g \cdot \overline{g}^t = \mathrm{I}_n\}.$$

**Lemma 5.5.** *The groups $\mathrm{O}(n), \mathrm{U}(n), \mathrm{Sp}(2n)$ are compact.*

*Proof.* Each column of a matrix in $\mathrm{O}(n)$ or $\mathrm{U}(n)$ or $\mathrm{Sp}(2n)$ is a unit vector, hence $\mathrm{O}(n)$ (resp. $\mathrm{U}(n), \mathrm{Sp}(2n)$) is a closed and bounded set in $\mathrm{Mat}_n(\mathbb{R})$ (resp. $\mathrm{Mat}_n(\mathbb{C}), \mathrm{Mat}_n(\mathbb{H})$). Therefore they are compact. □

**Lemma 5.6.** $\mathrm{Sp}(2n) \overset{\iota}{\simeq} \mathrm{U}(2n) \cap \mathrm{Sp}_{2n}(\mathbb{C})$.

*Proof.* In Example 3.10, we have two realizations of the complex symplectic groups $\mathrm{Sp}_{2n}(\mathbb{C})$: one via the form $\begin{pmatrix} & J_n \\ -J_n & \end{pmatrix}$, one via the form $\begin{pmatrix} & \mathrm{I}_n \\ -\mathrm{I}_n & \end{pmatrix}$. Two realizations are related by an isomorphism via the matrix $\begin{pmatrix} \mathrm{I}_n & \\ & J_n \end{pmatrix} \in \mathrm{U}(2n)$, thus $\mathrm{Sp}_{2n}(\mathbb{C}) \cap \mathrm{U}(2n)$ is independent of the choice of the above two forms.

It is easy to check for every $Z \in \mathrm{Mat}_n(\mathbb{H})$, $\iota(\overline{Z}^t) = \overline{\iota(Z)^t}$. Thus,

$$\begin{aligned}
&\iota(\mathrm{Sp}(2n)) = \iota(\mathrm{Sp}(2n) \cap \mathrm{Mat}_n(\mathbb{H})) \\
=&\left\{ g \in \mathrm{GL}_{2n}(\mathbb{C}) \,\middle|\, g \cdot \overline{g}^t = \mathrm{I}_{2n} \right\} \bigcap \left\{ g \in \mathrm{Mat}_{2n}(\mathbb{C}) \,\middle|\, g \begin{pmatrix} & -\mathrm{I}_n \\ \mathrm{I}_n & \end{pmatrix} = \begin{pmatrix} & -\mathrm{I}_n \\ \mathrm{I}_n & \end{pmatrix} \overline{g} \right\} \\
=&\mathrm{U}(2n) \bigcap \left\{ g \in \mathrm{GL}_{2n}(\mathbb{C}) \,\middle|\, g \begin{pmatrix} & -\mathrm{I}_n \\ \mathrm{I}_n & \end{pmatrix} g^t = \begin{pmatrix} & -\mathrm{I}_n \\ \mathrm{I}_n & \end{pmatrix} \right\} \\
=&\mathrm{U}(2n) \cap \mathrm{Sp}_{2n}(\mathbb{C}).
\end{aligned}$$

□

From now on, we will feel free to identify $\mathrm{Sp}(2n)$ with $\mathrm{Sp}_{2n}(\mathbb{C}) \cap \mathrm{U}(2n)$ at our convenience.

## 6. Homogeneous Space, $\pi_0$ and $\pi_1$ for Classical Compact Groups

To start with,

**Proposition 6.1.**

$$\mathrm{SU}(2) = \left\{ g = \begin{pmatrix} a & b \\ -\overline{b} & \overline{a} \end{pmatrix} \,\middle|\, |a|^2 + |b|^2 = 1, a, b \in \mathbb{C} \right\} \simeq S^3 (\textit{as smooth manifolds}).$$

*In particular,* $\pi_0(\mathrm{SU}(2)) = 0$, $\pi_1(\mathrm{SU}(2)) = 0$.

*Proof.* Recall that

$$\mathrm{SU}(2) = \{ g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid g \cdot \overline{g}^t = \mathrm{I}_2 \}.$$

This leads to the system of equations

$$|a|^2 + |b^2| = |c|^2 + |d|^2 = ad - bc = 1, \quad a\overline{c} + b\overline{d} = 0.$$

Set

$$a = \sin\alpha \cdot e^{\mathbf{i}\eta}, \quad b = \cos\alpha \cdot e^{\mathbf{i}\xi}, \quad c = \cos\beta \cdot e^{\mathbf{i}\gamma}, \quad d = \sin\beta \cdot e^{\mathbf{i}\delta}.$$

Here $\alpha, \beta \in [0, \frac{\pi}{2}], \eta, \xi, \gamma, \delta \in [0, 2\pi)$. The equation $a\overline{c} + b\overline{d} = 0$ can be rewritten as

$$\sin\alpha\cos\beta \cdot e^{\mathbf{i}(\eta-\gamma)} = \sin\beta\cos\alpha \cdot (-e^{\mathbf{i}(\xi-\delta)}).$$

By comparing the radius and the argument, we see that $\alpha = \beta$ and $e^{\mathbf{i}(\eta-\gamma)} + e^{\mathbf{i}(\xi-\delta)} = 0$. Then

$$1 = ad - bc = \sin^2\alpha \cdot e^{\mathbf{i}(\eta+\delta)} - \cos^2\alpha \cdot e^{\mathbf{i}(\xi+\gamma)} \leq \sin^2\alpha + \cos^2\alpha = 1.$$

The equality holds only when

$$e^{\mathbf{i}\eta} = e^{-\mathbf{i}\delta}, \qquad e^{\mathbf{i}(\xi+\gamma)} = -1.$$

This shows that $a = \overline{d}, c = -\overline{b}$. □

**Corollary 6.2.** *The group* $\mathrm{Sp}(2) \simeq \mathrm{SU}(2)$ *is connected and simply connected.*

*Proof.* Note that

$$\begin{aligned}\mathrm{Sp}_2(\mathbb{C}) =& \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{C}) \,\middle|\, \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} & 1 \\ -1 & \end{pmatrix} \begin{pmatrix} a & c \\ b & d \end{pmatrix} = \begin{pmatrix} & 1 \\ -1 & \end{pmatrix} \right\} \\ =& \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{C}) \,\middle|\, ad - bc = 1 \right\} = \mathrm{SL}_2(\mathbb{C}).\end{aligned}$$

Thus, $\mathrm{Sp}(2) = \mathrm{U}(2) \cap \mathrm{Sp}_2(\mathbb{C}) = \mathrm{U}(2) \cap \mathrm{SL}_2(\mathbb{C}) = \mathrm{SU}(2)$. □

Now we assume that $G$ is a Lie group and $H$ is a closed subgroup of $G$. We need the following theorem from differential geometry.

**Theorem 6.3** ([FWar83, Theorem 3.58])**.** *Let $\pi : G \to G/H$ be the natural projection. Then $G/H$ admits a unique smooth manifold structure such that*

(1) *$\pi$ is smooth*
(2) *there exists local smooth sections of $G/H$ in $G$: that is, if $\overline{g} \in G/H$, then there is a neighbourhood $W$ of $\overline{g}$ and a smooth map $s : W \to G$ with $\pi \circ s = \mathrm{Id}$.*

*Once $G/H$ is equipped with the unique smooth manifold structure as the above, then we call it a* ***homogeneous space***

A fancier way to address Theorem 6.3 is that $G$ **is a smooth principal $H$-bundle over the homogeneous space** $G/H$. Lots of smooth manifolds with certain symmetry arises in this way, hence can be studied via Lie theory; on the other hand, some topological properties of Lie groups can be studied via principal bundles by *induction*, which will be illustrated below.

Consider the principal $H$-bundle $H \to G \to G/H$. It is well-known that a principal bundle is a fibration (in the sense of Serre), and hence satisfies long exact sequence of homotopy groups.

**Theorem 6.4** ([Mi-To91, Chapter 2 Corollary 3.11])**.** *Assume that $G/H$ is connected. Then the principal $H$-bundle satisfies the following long exact sequence of homotopy groups:*

$$\cdots \to \pi_n(H) \to \pi_n(G) \to \pi_n(G/H) \to \pi_{n-1}(H) \to \cdots \to \pi_0(H) \to \pi_0(G) \to 0.$$

**Remark 6.5.** *The additional assumption that $G/H$ is connected comes from my personal laziness, in order to avoid further technical explanations on $\pi_0$.*

We will for sure take the above theorem for granted. There is no way to cover its proof in a course on representation theory. Anyway, we only need some low degree homotopy groups in the above exact sequence.

**Corollary 6.6** ([FWar83, Proposition 3.66])**.** *If $H$ is connected, $G/H$ is connected, then $G$ is connected.*

*Proof.* The corollary is obvious from the last terms of the long exact sequence of homotopy groups:

$$\pi_0(H) \to \pi_0(G) \to 0.$$

□

The computation of $\pi_1(G)$ is slightly more complicated, as it is inevitable to involve some second homotopy groups $\pi_2(G/H)$. In fact, for classical groups, we only need the following result.

**Lemma 6.7.** *Assume $0 < i < n$. Then $\pi_i(S^n) = 0$. In particular, for $n \geq 3$, $\pi_2(S^n) = 0$.*

**Theorem 6.8.** *For $n \geq 2$, the group $\mathrm{SU}(n)$ is connected and simply connected.*

*Proof.* Realize $\mathbb{C}^n$ as the space of column vectors. Consider the natural action of $\mathrm{SU}(n)$ on $\mathbb{C}^n$ by matrix multiplication on the left. As $\mathrm{SU}(n)$ preserves the standard inner product on $\mathbb{C}^n$, it induces a transitive action of $\mathrm{SU}(n)$ on the unit sphere of $\mathbb{C}^n$, i.e. $S^{2n-1}$. The stabilizer of this action is $\mathrm{SU}(n-1)$. Thus we have a principal bundle:

$$\mathrm{SU}(n-1) \to \mathrm{SU}(n) \to S^{2n-1}.$$

A simple induction via Corollary 6.6 shows that all special unitary groups $\mathrm{SU}(n)$ $(n \geq 2)$ are connected.

By Theorem 6.4, we have a long exact sequence of homotopy groups

$$\pi_2(S^{2n-1}) \to \pi_1(\mathrm{SU}(n-1)) \to \pi_1(\mathrm{SU}(n)) \to \pi_1(S^{2n-1}).$$

As $\pi_2(S^{2n-1}) = 0$ (Lemma 6.7) and $\pi_1(S^{2n-1}) = 0$, an easy induction shows that for $n \geq 2$

$$\pi_1(\mathrm{SU}(n+1)) \simeq \pi_1(\mathrm{SU}(n)) \simeq \cdots \simeq \pi_1(\mathrm{SU}(2)) = 0.$$

Thus all special unitary groups $\mathrm{SU}(n)$ $(n \geq 2)$ are simply connected. □

**Theorem 6.9.** (1) *For $n \geq 2$, $\mathrm{SO}(n)$ is connected.*
(2) $\pi_1(\mathrm{SO}(2)) \simeq \mathbb{Z}$
(3) *For $n \geq 3$, $\pi_1(\mathrm{SO}(n)) \simeq \mathbb{Z}/2\mathbb{Z}$.*

*Proof.* The proof is similar to that of Theorem 6.8. We have a principal bundle

$$\mathrm{SO}(n-1) \to \mathrm{SO}(n) \to S^{n-1}.$$

Then a simple induction via Corollary 6.6 shows that all $\mathrm{SO}(n)$ $(n \geq 2)$ are connected. It is obvious that $\pi_1(\mathrm{SO}(2)) \simeq \pi_1(S^1) \simeq \mathbb{Z}$.

The long exact sequence of homotopy groups implies that for $n \geq 3$

$$\pi_1(\mathrm{SO}(n+1)) \simeq \pi_1(\mathrm{SO}(n)) \simeq \cdots \simeq \pi_1(\mathrm{SO}(3)).$$

It remains to compute $\pi_1(\mathrm{SO}(3))$. For this part, we only need to show that $\mathrm{SU}(2)$ is a double cover of $\mathrm{SO}(3)$. We leave this part to the next lecture. □

**Theorem 6.10.** *For $n \geq 2$, the group $\mathrm{Sp}(2n) = \mathrm{U}(2n) \cap \mathrm{Sp}_{2n}(\mathbb{C})$ is connected and simply connected.*

*Proof.* By identifying $\mathrm{Sp}(2n)$ as the isometry group of the norm $|x| = \sqrt{x \cdot \overline{x}}$ on $\mathbb{H}^n \simeq \mathbb{R}^{4n}$ (as real vector spaces), we get a principal bundle:

$$\mathrm{Sp}(2n-2) \to \mathrm{Sp}(2n) \to S^{4n-1}.$$

A simple induction via Corollary 6.6 shows that all compact symplectic groups $\mathrm{Sp}(2n)$ $(n \geq 1)$ are connected.

By Theorem 6.4, we have a long exact sequence of homotopy groups

$$\pi_2(S^{4n-1}) \to \pi_1(\mathrm{Sp}(2n-2)) \to \pi_1(\mathrm{Sp}(2n)) \to \pi_1(S^{4n-1}).$$

As $\pi_2(S^{4n-1}) = 0$ (Lemma 6.7) and $\pi_1(S^{4n-1}) = 0$, an easy induction shows that for $n \geq 1$

$$\pi_1(\mathrm{SU}(n+1)) \simeq \pi_1(\mathrm{SU}(n)) \simeq \cdots \simeq \pi_1(\mathrm{SU}(2)) = 0.$$

Thus all compact symplectic groups $\mathrm{Sp}(2n)$ $(n \geq 1)$ are simply connected. □

**Remark 6.11.** *Higher dimensional (up to a certain dimension) homotopy groups can be computed in the same way, for instance, see* [Mi-To91, Chapter 2 Corollary 3.17].

So far, we have computed $\pi_0$ and $\pi_1$ for compact classical groups. How about noncompact groups? For non-compact real reductive groups (though we haven't defined them and maybe never will in this course), we have certain decomposition theorems implying that homotopic invariants of real reductive groups are the same as those of their maximal compact subgroups. Thus, all homotopic invariants are known once we know their computation for compact Lie groups.

**Example 6.12.** *The group* $\mathrm{SO}_n(\mathbb{C})$ *($n \geq 3$) is non-compact, its maximal compact subgroup is* $\mathrm{SO}_n(\mathbb{C}) \cap \mathrm{U}(n) = \mathrm{SO}(n)$. *Thus* $\mathrm{SO}_n(\mathbb{C})$ *is connected, but not simply connected.* $\pi_1(\mathrm{SO}_n(\mathbb{C})) \simeq \mathbb{Z}/2\mathbb{Z}$ *($n \geq 3$).*

We end this section by pointing out the following theorem on the lifting of Lie algebra homomorphism.

**Theorem 6.13** ([FWar83, Theorem 3.27])**.** *Let $G, H$ be Lie groups and $\mathfrak{g}, \mathfrak{h}$ be their Lie algebras. We further assume that $G$ is simply connected, then for every Lie algebra homomorphism $\psi : \mathfrak{g} \to \mathfrak{h}$, there exists a Lie group homomorphism $\varphi : G \to H$ such that $\psi = d\varphi$. In particular, if two simply connected Lie groups have isomorphism Lie algebras, then they are isomorphic.*

*Proof.* The uniqueness is addressed in Proposition 4.7; while the existence relies on the theory of differential ideals and integral submanifolds. We omit it. □

A special case is that $\psi : \mathfrak{g} \to \mathfrak{h}$ is an isomorphism of Lie algebras. Then the lifting $\varphi : G \to H$ is a covering homomorphism. We will explain this in the next lecture.

## Appendix A. Several Proofs regarding Topological Groups

Theorem 1.9 follows from Baire Category theorem and Open Mapping Theorem for topological groups.

**Theorem A.1.** *Every locally compact Hausdorff space is a Baire space, meaning that every countable intersection of open dense subsets is still dense.*

**Theorem A.2.** *Let $G, H$ be locally compact groups and $\pi : G \to H$ be a surjective homomorphism. Under a further assumption that $G$ is $\sigma$-compact, $\pi$ is open.*

*Proof.* Let $U$ be an open neighbourhood of $e_G$. Take a sufficiently small symmetric compact neighbourhood $V$ of $e_G$ such that $V^2 \subseteq U$. Since $G$ is $\sigma$-compact, we can take a sequence $x_n \in G$ such that $G = \bigcup_n x_n V$. Then $H = \pi(G) = \bigcup_n \pi(x_n)\pi(V)$. By Baire's theorem, there exists some $n$ such that $\pi(x_n)\pi(V)$ has non-empty interior. Via translate by $\pi(x_n)^{-1}$, $\pi(V)$ is has non-empty interior.

Say $W$ is the interior of $\pi(V)$. Note that $V = V^{-1}$, we have $W = W^{-1}$. We check that $W^2$ is an open neighbourhood of $e_H$ contained in $\pi(U)$. Indeed, if $y \in W$, then $y^{-1} \in W^{-1} = W$, hence $e_H = y \cdot y^{-1} \in W^2 \subseteq \pi(V^2) \subseteq \pi(U)$. Hence $\pi$ is open. □

Next we prove Corollary 1.14.

*Proof of Corollary 1.14.* By Lemma 1.13, the intersection of all neighbourhoods of $e$ is equal to the intersection of all closed symmetric neighbourhoods of $e$, hence it contains $\overline{\{e\}}$.

If there were $x \in G$ in the intersection of all neighbourhoods of $e$ that is not in $\overline{\{e\}}$, then we would find an open neighbourhood $U$ of $e$ such that $e \notin xU$. By Lemma 1.13, we take a closed symmetric neighbourhood $K$ of $\{e\}$ contained in $U$. Then $e \notin xK$, or equivalently, $x^{-1} \notin K$. This leads to the contradiction that $x \notin K$.

By Lemma 1.12, $\overline{\{e\}}$ is a closed subgroup of $G$. If $H$ is a closed subgroup of $G$, then $e \in H$ hence $\overline{\{e\}} \subseteq H$. To check normality of $\{e\}$, we notice that for every $g \in G$, $g \cdot \overline{\{e\}} \cdot g^{-1}$ is a closed subgroup of $G$ containing $e$, hence $\overline{\{e\}} \subseteq g \cdot \overline{\{e\}} \cdot g^{-1}$. Then the normality follows. □

## Appendix B. Function Spaces and Measures

In this section, we always assume that $X$ is a locally compact Hausdorff topological space. As usual, we write $C(X)$ for the space of $\mathbb{C}$-valued continuous functions, and write $C_c(X)$ for the space of $\mathbb{C}$-valued continuous functions with compact support. Here we recall that the support of a function

$$\operatorname{supp}(f) := \text{ the closure of } \{x \in X \mid f(x) \neq 0\}$$

is always a closed subset. If $K$ is a compact subset of $X$, then we write $C_K(X)$ for the space of $\mathbb{C}$-valued functions with support in $K$. These function spaces are obviously $\mathbb{C}$-vector spaces. Hence throughout this section, unless emphasized otherwise, linearity always refers to $\mathbb{C}$-linearity. Naturally, we have a linear injection:

$$C_K(X) \hookrightarrow C_c(X) \hookrightarrow C(X).$$

Now we need to equip a topology on $C_c(X)$. Define a norm on $C_c(X)$

$$\|f\|_\infty := \sup\{|f(x)| \mid x \in X\}.$$

Unfortunately $C_c(X)$ is not *in general* a Banach space under $\|\cdot\|_\infty$, as this norm is not complete (yet when $X$ is compact, $C_c(X) = C(X)$ is a Banach space). Define $C_0(X)$ to be the completion of $C_c(X)$ under $\|\cdot\|_\infty$. This space is called the space of **functions vanishing at infinity**.

**Lemma B.1.** *For any compact subset $K$, $C_K(X)$ is a closed subspace of $C_0(X)$. Hence $C_K(X)$ is a Banach space under $\|\cdot\|_\infty$.*

*Proof.* This is trivial. □

The following topological lemma is well-known.

**Lemma B.2.** *There exists a sequence of compact subsets $K_j$ exhausting $X$, i.e. $K_1 \subseteq K_2 \subseteq K_3 \cdots$ and $\bigcup_j K_j = X$.*

Then we have an injection of Banach spaces:

$$C_{K_1}(X) \hookrightarrow C_{K_2}(X) \hookrightarrow C_{K_3}(X) \cdots$$

and $C_c(X)$ is regarded as the inductive limit of the topological vector spaces $C_{K_j}(X)$ (such spaces are called LF-spaces, limit of Fréchet spaces in general). The point is that it's better to use the inductive limit topology on $C_c(X)$ (not the original norm $\|\cdot\|_\infty$).

**Lemma B.3.** (1) *Once equipped with the inductive limit topology, $C_c(X)$ is a complete locally convex topological vector space.*

(2) *For any compact subset $K$, the Banach space $C_K(X)$ topologically embeds into $C_c(X)$.*
(3) *A linear functional $\mu$ is continuous on $C_c(X)$ if and only if its restriction on any $C_K(X)$ is continuous.*

*Proof.* This follows from standard functional analysis. For details, see [Tr67, Chapter 13]. □

**Corollary B.4.** *A linear functional $\mu$ on $C_c(X)$ is continuous if and only if for every compact subset $K \subseteq X$, there exists a constant $M_K$ such that*

$$|\mu(f)| \leq M_K \|f\|_\infty$$

*for all $f \in C_K(X)$.*

*Proof.* This is just a translation of Lemma B.3. □

**Definition B.5.** *A (Radon)* ***measure*** *$\mu$ on $X$ is defined to be a continuous linear functional on $C_c(X)$. In analogy with the classical notation, we also write $\mu(f)$ as $\int_X f(x)d\mu(x)$ or $\int_X f d\mu$*

**Definition B.6.** *A function $f \in C_c(X)$ is said to be* ***positive*** *if $f$ takes value in $\mathbb{R}$ and $f(x) \geq 0$ for all $x \in X$. A measure is said to be* ***positive*** *if for any positive function $f \in C_c(X)$, $\mu(f) \geq 0$.*

**Remark B.7.** *The zero measure $\mu = 0$ is positive by definition. Yet this may include some technical pathological examples. For instance, one may attempt to define the $L^p$-space to be the completion of $C_c(X)$ under the $L^p$-'norm' $\|f\|_p := (\int_X |f|^p d\mu)^{\frac{1}{p}}$. Yet this definition requires more conditions on $\mu$, for instance, the support of $\mu$ should be large enough.*

*Think about $\mathrm{O}(2) \simeq \mathrm{SO}(2) \times \mathbb{Z}/2\mathbb{Z}$ as manifolds. We equip the usual invariant measure $d\theta$ on the identity component of $\mathrm{O}(2)$ and equip the zero measure on the other component, i.e.*

$$\int_{\mathrm{O}(2)} f d\mu := \int_{\mathrm{SO}(2)} f(e^{\mathbf{i}\theta}) d\theta = \int_0^{2\pi} f(e^{\mathbf{i}\theta}) d\theta,$$

*then $f \mapsto \int_{\mathrm{O}(2)} |f| d\mu$ is not a norm.*

*Yet if we avoid this kind of bad phenomena, $L^p$-spaces can always be regarded as completions of $C_c(X)$.*

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